AI-Productivity-Assistant

An AI-powered workplace productivity assistant that automates common office tasks: email generation, meeting summarization, task planning, research assistance, and chatbot interaction.

Built to demonstrate effective AI tool usage, prompt engineering, and responsible AI practices.

🚀 Live Demo

Open `index.html` in your browser for the full UI, or run `app.py` for the Python backend.

✨ Features

1. **Email Generation:** Draft professional emails from brief prompts. Tone control (formal, friendly, persuasive).

2. **Meeting Summarization:** Paste transcripts or notes → Get summary, action items, decisions, and owners.

3. **Task Planning:** Breaks a goal into SMART tasks, prioritizes with Eisenhower Matrix, and creates a daily schedule.

4. **Research Assistance:** Structured research briefs with sources, key insights, and counter-arguments.

5. **Chatbot Interaction:** Context-aware workplace Q&A assistant.

🛠️ Tech Stack

- **Frontend:** HTML5, CSS3, JavaScript (Vanilla) - `index.html`
- **Backend (Optional):** Python Flask - `app.py`
- **AI Integration:** OpenAI API / Gemini API / HuggingFace (pluggable) - configured in `config.py`
- **Prompt Engineering:** Library of tested prompts in `prompts.py`

🗂️ Project Structure

```
AI-Productivity-Assistant/
├── index.html          # Main web app - All-in-one functional UI
├── app.py              # Flask backend API (optional deployment)
├── prompts.py          # Prompt engineering library
├── config.py           # AI API configuration
├── requirements.txt
├── README.md
├── .gitignore
└── docs/
    ├── ETHICS.md       # Responsible AI use
    └── PROMPT_GUIDE.md # How prompts were engineered
```

Getting Started

1. Create Your GitHub Profile

- Go to https://github.com/signup
- Use your personal email
- Add profile photo, bio, and pin this repo

2. Set Up Repository

```
git init
git add .
git commit -m "Initial commit: AI Productivity Assistant"
# Create repo on GitHub named AI-Productivity-Assistant
git remote add origin https://github.com/YOUR_USERNAME/AI-Productivity-Assist
git branch -M main
git push -u origin main
```

3. Run Locally

Option A: Just Frontend (No setup needed)
Simply open `index.html` in any browser. It works offline with mock AI responses and you can plug in your API key in the UI.

Option B: Full Stack

```
pip install -r requirements.txt
python app.py
# Visit http://localhost:5000
```

Set your API key in `config.py` or as environment variable:

```
OPENAI_API_KEY=sk-...
```

🧠 Prompt Engineering Highlights

See `prompts.py` and `docs/PROMPT_GUIDE.md`. Techniques used:

- Role-based prompting ("You are a senior executive assistant...")
- Chain-of-Thought for task planning
- Few-shot examples for email tone
- Structured output (JSON) for meeting summaries
- Self-critique loop for research

⚖️ Responsible & Ethical Use

- No data is stored without consent
- Transparency: AI-generated content is labeled
- Privacy: Meeting transcripts are processed locally first
- Bias check: Prompts tested for neutral tone

See `docs/ETHICS.md` for full statement.

📸 Screenshots

Add screenshots of your running app here for your submission.

Future Improvements

- Integration with Gmail / Google Calendar / Notion API

- Voice-to-text meeting capture
- Team collaboration mode

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